The documentation and process conversion measures necessary to comply with this revision shall be completed by 30 June 2003.

INCH-POUND

MIL-PRF-19500/427J
30 December 2002
SUPERSEDING
MIL-PRF-19500/427H
8 July 2000

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER,
TYPES 1N5614, 1N5616, 1N5618, 1N5620, 1N5622, 1N5614EG1, 1N5616EG1, 1N5618EG1, 1N5620EG1, 1N5622EG1, 1N5614UL, 1N5616UL, 1N5618UL, 1N5620UL, 1N5622UL, 1N5614US, 1N5616US, 1N5618US, 1N5620US, 1N5622US, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, hermetically sealed power rectifier diodes. Four levels of product assurance are provided for each encapsulated device as specified in MIL-PRF-19500. Two levels of product assurance are provided each unencapsulated device type.

1.2 Physical dimensions. See figures 1, (similar to DO-204; formerly DO-41), 2, 3 (surface mount), 4, and 5 (JANHC and JANKC).

* 1.3 Maximum ratings.

Col. 1	Col. 2	Col. 3	Col. 4		Col. 5		Col. 6	Col. 7	Col. 8	Col. 9	Col. 10
Types	V_R	V_{RWM}	I_O		I_{FSM} T_A = +100°C I_F = 750 mA t_p = 8.3 ms		T_J (4)	t_{rr}	$R_{\theta JL}$ (5)	$Z_{\theta JX}$	T_{STG}
			T_A = +55°C (1) (2)	T_A = +100°C (2) (3)		EG1 suffix					
	V dc	V(pk)	A	mA	A(pk)	A(pk)	°C	µs			°C
1N5614, UL, US, EG1	200	200	1	750	30	25	-65	2	See figure 6	4.5 °C/W	0
1N5616, UL, US, EG1 (6)	400	400	1	750	30	25		2			
1N5618, UL, US, EG1 (6)	600	600	1	750	30	25	to	2			to
1N5620, UL, US, EG1 (6)	800	800	1	750	30	25		2			
1N5622, UL, US, EG1 (6)	1,000	1,000	1	750	30	25	+200	2			+175

(1) From 1 A at T_A = +55°C, to 0.75 A at T_A = +100°C, derate linearly at 5.56 mA/°C.
(2) For the 1 A rating at the +55°C and the 750 mA rating at +100°C, no special mounting, heat sinking, or forced air flow across exposed areas of the device is required.
(3) From 0.75 A at T_A = +100°C, to 0 A at T_A = +200°C, derate linearly at 10 mA/°C.
(4) For EG1 parts T_J and T_{OP} = 175°C max.
(5) $R_{\theta JL} \leq$ 38°C/W at L = .375 inch (9.53 mm) for UL and US suffix type devices, $R_{\theta JEC}$ = 7.0°C/W.
(6) Barometric pressure reduced: 1N5616 and 1N5618 = 8 mm Hg, 1N5620 and 1N5622 = 33 mm Hg.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, Post Office Box 3990, Columbus, OH 43213-1999, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

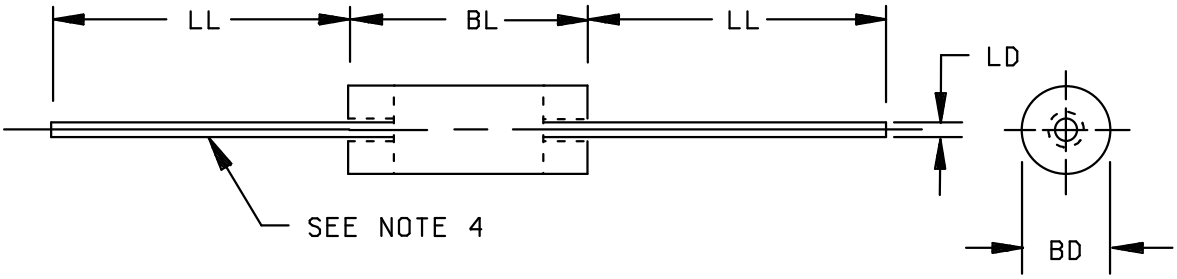
3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

* 3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

EC	End cap.
EG	Epoxy over glass hermetically sealed EG1 case (see figure 6).
UL	Long-body unleaded or surface mounted diodes (square end-caps).
US	Short-body unleaded or surface mounted diodes (square end-caps).

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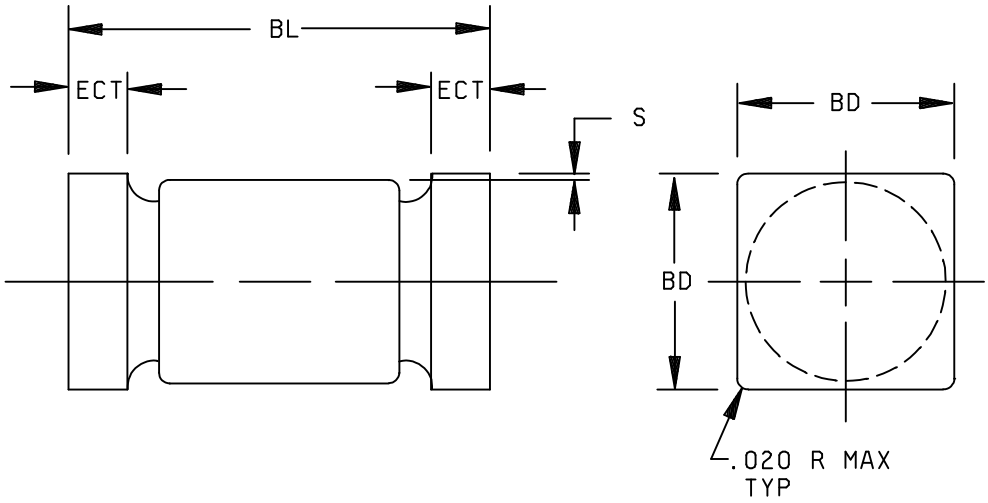


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BD	.065	.110	1.65	2.79	3
BL	.130	.225	3.30	5.72	4
LD	.026	.033	0.66	0.84	
LL	1.00	1.30	25.4	33.02	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension BD shall be measured at the largest diameter.
4. The BL dimension shall include all uncontrolled areas of the device.
5. The shape of the body, within the bounds of the dimensions is optional.

FIGURE 1. Physical dimensions for axial leaded devices only (similar to DO-204, formerly DO-41).

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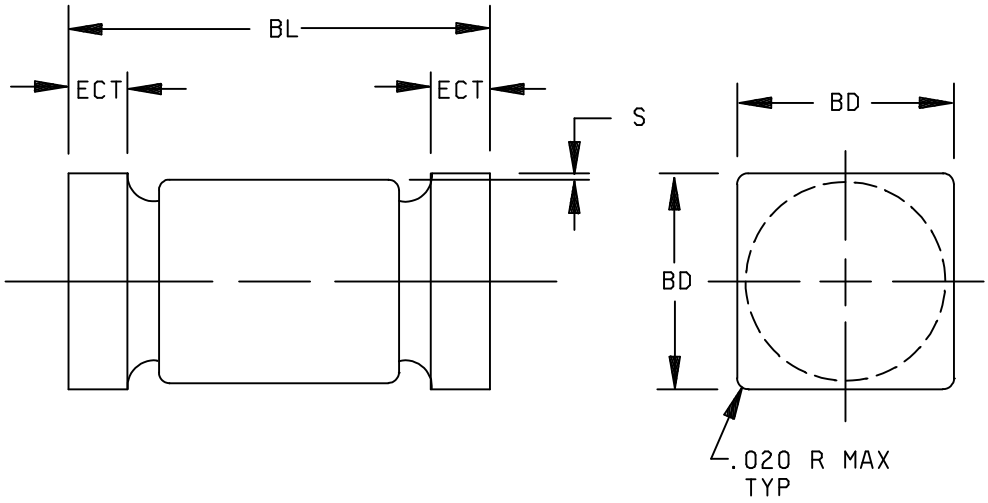


Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BD	.091	.125	2.31	3.18
BL	.168	.225	4.27	5.72
ECT	.019	.028	0.48	0.71
S	.003		0.08	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions types 1N5614UL, 1N5616UL, 1N5618UL, 1N5620UL, and 1N5622UL (surface mount devices).

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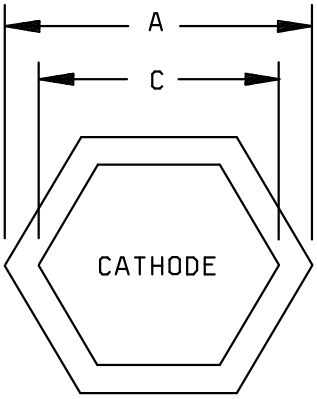


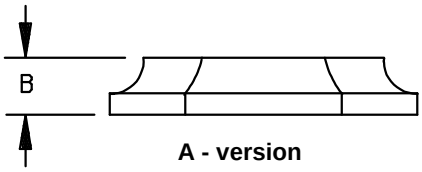
Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BD	.091	.103	2.31	2.62
BL	.168	.200	4.27	5.08
ECT	.019	.028	0.48	0.71
S	.003		0.08	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. Physical dimensions types 1N5614US, 1N5616US, 1N5618US, 1N5620US, and 1N5622US (surface mount devices).

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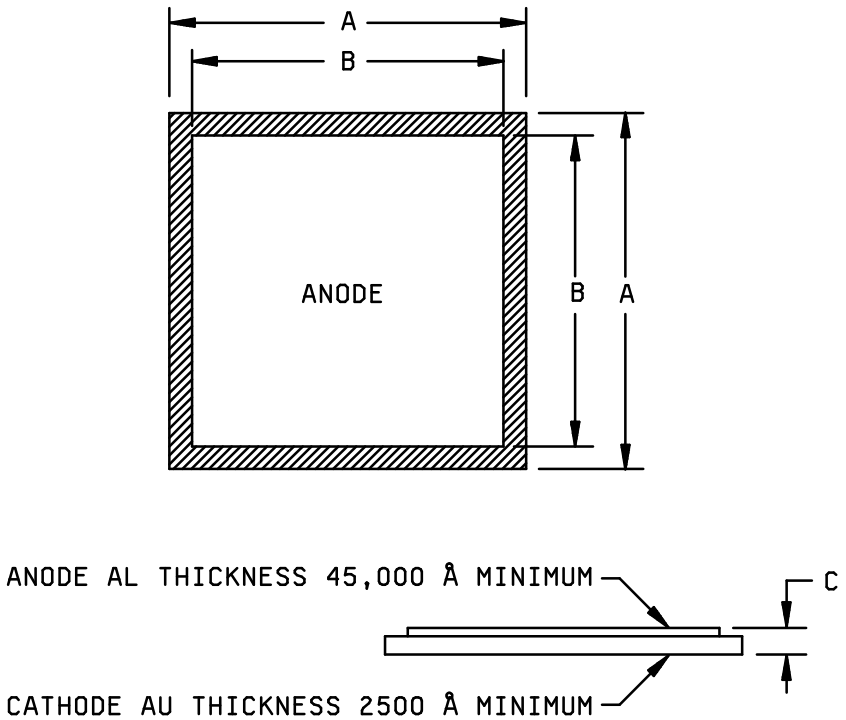
A - version

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.047	.053	1.19	1.35
B	.007	.011	0.18	0.28
C	.033	.037	0.84	0.94

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:
   Top metal: Gold 10,000 Å minimum.
   Back metal: Gold 4,000 Å minimum.

FIGURE 4. Physical dimension, JANHCA AND JANKCA die.

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B - version

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.039	.043	1.00	1.09
B	.021	.025	0.53	0.64
C	.008	.012	0.20	0.31

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:
   Top metal: Aluminum (anode) . . . . . 45,000 Å minimum.
   Back metal: Gold (cathode) . . . . . 2,500 Å minimum.

FIGURE 5. Physical dimension, JANHCB AND JANKCB die.

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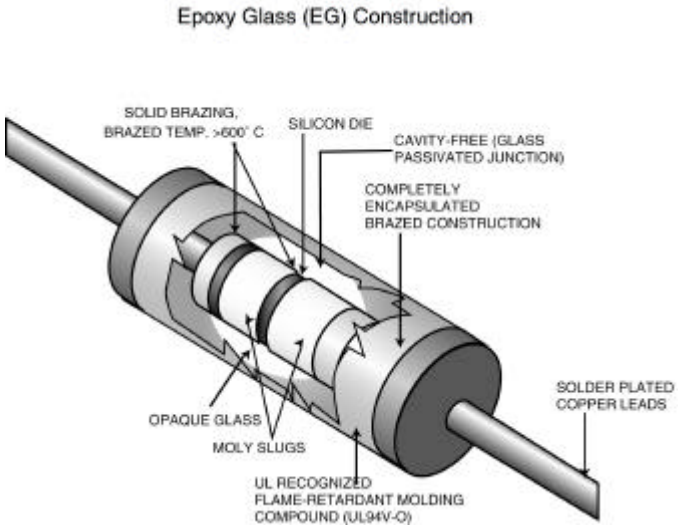


Figure 6 Epoxy-Glass (EG1 case outline) construction (for reference only).

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3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and in figures 1, 2, 3, 4, and 5 herein. Except for JANHC and JANKC, all devices shall be metallurgically bonded-thermally-matched-noncavity-double plug construction, utilizing a category I bond, in accordance with MIL-PRF-19500. UL and US version devices shall be structurally identical to the nonsurface mount version devices except for lead configuration.

* 3.4.1 Lead finish. Unless otherwise specified, lead or end cap finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. When solder alloy is used for finish, the maximum lead temperature is limited to 175°C maximum. Where a choice of finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5.1 Marking of UL and US version devices. For UL and US version devices only, all marking (except as stated in 3.6) may be omitted from the body, but shall be retained on the initial container.

* 3.5.2 Marking for EG1 version. The EG1 version, (epoxy over glass construction), in this associated specification is a construction identifier. It denotes a type of design and not a change in package outline. EG1 is not intended to be a new part number or a new package outline identifier. The EG1 version meets all of the requirements of a non-EG1 version including case outline, hermeticy and electrical parameters. The EG1 may be marked and shipped as a non-EG1 part number because it is an equivalent device in terms of form, fit and function. The EG1 is interchangeable and substitutable for the non-EG1 version.

3.6 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for UL and US suffix devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end or a contrasting colorband may be used.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and table I herein.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I, herein.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

* 4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.2.1 Group E qualification. Group E qualification shall be performed herein for qualification or requalification only. In case qualification was awarded to a prior revision of the associated specification that did not request the performance of table II tests, the tests specified in table II herein shall be performed by the first inspection lot to this revision to maintain qualification.

4.2.2 JANHC and JANKC die. Qualification shall be in accordance with appendix H of MIL-PRF-19500.

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4.3 Screening (JANTX, JANTXV and JANS levels only). For appendix D qualified suppliers, screening shall be in accordance with table IV of MIL-PRF-19500, appendix E, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable. For appendix C qualified suppliers (QML), refer to QM plan for screening requirements.

Screen (see, table IV of MIL-PRF-19500, appendix E)	Measurement	
	JANS level	JANTX and JANTXV levels
(1)	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
9	I_{R1} and V_F	Not applicable
11	I_{R1} and V_F, $\Delta I_{R1} \leq 100$ percent of initial reading or $\pm$ 100 nA dc, whichever is greater. $\Delta V_F \leq \pm$ 0.1 V dc.	I_{R1} and V_F
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein: $\Delta I_{R1} \leq 100$ percent of initial reading or $\pm$ 100 nA dc, whichever is greater. $\Delta V_F \leq \pm$ 0.1 V dc. Scope display evaluation (see 4.5.4)	Subgroup 2 of table I herein: $\Delta I_{R1} \leq 100$ percent of initial reading or $\pm$ 100 nA dc, whichever is greater. $\Delta V_F \leq \pm$ 0.1 V dc. Scope display evaluation (see 4.5.4)

(1) Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.

* 4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: Method 1038 of MIL-STD-750, condition B, V_R = full rated V_{RWM} (see 1.3); f = 50-60 Hz, I_O = 1 A dc minimum, (see 4.5.3 and 4.5.3.1).

- Burn-in set up. Place a single device in your burn-in/life test rack. Adjust power (or current) until the part reaches T_J = 135°C minimum. This is the minimum power/current which must be applied to each device. Use method 3100 to measure T_J. Additional heat generated by self-heating is allowed but shall not cause any device to exceed the maximum temperature rating.

4.3.2 Screening (JANHC and JANKC). Screening of die shall be in accordance with appendix H of MIL-PRF-19500. As a minimum, die shall be 100-percent probed to ensure compliance with group A, subgroup 2 of MIL-PRF-19500. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

* 4.3.3 Thermal impedance $Z_{\theta JX}$ measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3101 of MIL-STD-750. The maximum screen limit shall be developed by the supplier using statistical methods and it shall not exceed the table I, subgroup 2 herein.

* 4.3.3.1 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3101 of MIL-STD-750 (read and record date $Z_{\theta JX}$). $Z_{\theta JX}$ shall be supplied on one lot (500 pieces minimum and a thermal response curve shall be submitted.) Twenty-two of these samples shall be serialized and provided to the qualifying activity for correlation prior to shipment of parts. Measurements conditions shall be in accordance with 4.4.1.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed in accordance with E.5.3.1d of MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

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* 4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table V of MIL-PRF-19500, appendix E, and table I herein. $Z_{\theta JX}$ measurements in table I, subgroup 2 shall be performed in accordance with method 3101 of MIL-STD-750.

a. I_M measuring current 10 mA.

b. I_H forward heating current 10 A.

c. t_H heating time 10 ms.

d. t_{MD} measurement delay time 100 µs maximum.

The maximum limit for $Z_{\theta JX}$ in table I, subgroup 2 is $Z_{\theta JX}$ (maximum) = 4.5°C/W.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500 appendix E, and as follows. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, subgroup 2 herein. Delta measurements shall be as specified in table III herein. Leaded samples may be used in lieu of surface mount devices (UL and US version) for life tests.

* 4.4.2.1 Group B inspection table VIa (JANS) of MIL-PRF-19500 appendix E.

	Subgroup	Method	Conditions
*	B3	1056	0°C to +100°C, 10 cycles.
*	B3	1051	-55°C to +175°C, 20 cycles.
*	B3	4066	I_{FSM} = rated I_{FSM} (see col. 5 of 1.3); 10 surges of 8.3 ms each at 1 minute intervals, superimposed on I_O = 0, V_{RWM} = 0.
	B4	1037	I_O = I_O (rated) minimum; V_R = rated V_{RWM} (see 1.3).
	B5	1027	I_O = I_{O2} rated dc minimum (see 1.3, col. 4); apply V_R = rated V_{RWM} (see 1.3, col. 3 and 4.5.6) adjust I_O to achieve T_J minimum; f = 50-60 Hz.
		or	Option 1: T_A = + 30°C max. ; T_J = 225°C minimum; t = 216 hours; n = 45, c = 0.
			Option 2: T_A = + 75°C max. ; T_J = 275°C minimum; t = 96 hours.
	B6	3101 or 4081	+25°C ≤ T_A ≤ +35°C (recorded before test is performed); $R_{\theta JL}$ (maximum) ≤ 38°C/W; L = .375 inch (9.53 mm) (see figure 7). For surface mount devices (UL and US version), $R_{\theta JEC}$ ≤ 7°C/W.
	B7		Peak reverse power, see 4.5.5 and figure 8 herein. P_{RM} ≥ 500 W. Test shall be performed on each sublot; sampling plan n = 10, c = 0, end-points, see 4.4.2.

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4.4.2.2 Group B inspection table VIb (JANTX and JANTXV) of MIL-PRF-19500 appendix E.

	Subgroup	Method	Conditions
*	B2	1056	0°C to +100°C, 10 cycles.
*	B2	1051	-55°C to +175°C,20 cycles.
*	B3	1027	Adjust I_O to achieve the required T_J; f = 50-60 Hz; apply V_R = rated V_{RWM} (see 1.3 and 4.5.3.1).

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, appendix E and as follows. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, subgroup 2 herein. Delta measurements shall be as specified in table III herein. Leaded samples may be used in lieu of surface mount devices (UL and US version) for life tests.

4.4.3.1 Group C inspection table VII of MIL-PRF-19500 appendix E.

	Subgroup	Method	Conditions
*	C2	1056	0°C to +100°C, 10 cycles.
*	C2	1051	-55°C to +175°C,20 cycles.
	C2	2036	Tension: Test condition A, weight = 12 pounds. Lead fatigue: Test condition E, weight = 1 pound. NOTE: Neither tension nor lead fatigue are applicable to surface mount devices (UL and US version).
*	C6	1026	Adjust I_O to achieve the required T_J; f = 50-60 Hz; apply V_R = rated V_{RWM} (see 1.3 (col. 3), and 4.5.3.1).

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500, appendix E, and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein; except, $Z_{\theta JX}$ need not be performed. See table III for delta limits when applicable.

4.5 Methods of inspection. Methods of inspection shall be specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Inspection conditions. Unless otherwise specified, all inspections shall be conducted at an ambient temperature; T_A = room ambient.

4.5.3 Burn-in and life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180 degrees, nor less than 150 degrees.

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* 4.5.3.1 Delta T_J rise. Io is adjusted such that the junction rise above ambient shall be delta T_J = 110°C min. for each diode and the junction temperature of each diode is maintained at T_J = 135°C min for burn-in and T_J = 150°C min for life testing. The delta T_J rise is intended to be achieved by device current and not by external or self-heating. ($T_J - T_A$) $\geq$ 100°C Deliberate heat sinking or forced air-cooling is prohibited unless otherwise approved by the qualifying activity. Ambient temperature shall be controlled to prevent T_J from exceeding rated T_J. The use of a current limiting or ballast resistor is permitted provided that each DUT still sees the full P_t (minimum) and that the minimum applied voltage, where applicable, is maintained through out the burn-in period.

* 4.5.4 Scope display evaluation. Scope display evaluation shall be sharp and stable in accordance with method 4023 of MIL-STD-750. Scope display may be performed on ATE (automatic test equipment) for screening only with the approval of the qualifying activity. Scope display in table I, subgroup 4 shall be performed on a scope. Reverse current (I_{BR}) over the knee shall be 500 µA peak.

4.5.5 Peak reverse power test. This test shall be measured in the circuit of figure 7, or equivalent. A 20 microsecond half-sine waveform of current shall be used and peak reverse power shall be determined by the product of peak reverse voltage and peak reverse current.

4.5.6 Thermal resistance. Thermal resistance measurement shall be performed in accordance with method 3101 or 4081 of MIL-STD-750. Forced moving air or draft shall not be permitted across the device during test. The maximum limit for $R_{\theta JL}$ under these test conditions shall be $R_{\theta JL(max)}$ = 38°C/W at .375 inch (9.53 mm) and 7°C/W at 0 inch (0.00 mm).

4.5.7 Life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180 degrees nor less than 150 degrees.

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TABLE I. Group A inspection.

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions 1/		Min	Max	
* Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Forward voltage	4011	I_F = 3 A dc, pulsed (see 4.5.1) t_p = 300 µs, 2 maximum duty cycle	V_F	0.8	1.3	V dc
Reverse current leakage	4016	DC method; V_R = rated V_R (see 1.3)	I_{R1}		0.5	µA dc
Breakdown voltage	4021	I_R = 50 µA dc	$V_{(BR)R}$			
1N5614, UL, US, EG1				200		V dc
1N5616, UL, US, EG1				400		V dc
1N5618, UL, US, EG1				600		V dc
1N5620, UL, US, EG1				800		V dc
1N5622, UL, US, EG1				1,000		V dc
Thermal impedance	3101	See 4.3.3	$Z_{\theta JX}$		4.5	°C/W
Subgroup 3						
High temperature operation:		T_A = +100°C				
Reverse current leakage	4016	DC method; V_R = rated V_R (see 1.3)	I_{R2}		25	µA dc
Subgroup 4						
Reverse recovery time	4031	Condition B1	t_{rr}		2	µs
Scope display	4023	See 4.5.4; n = 116, c = 0				
Subgroup 5						
Not applicable						
Subgroup 6						
Forward surge	4066	I_{FSM} = 30 A (pk); 10 surges of 8.3 ms each at 1 minute intervals, superimposed on I_O = 750 mA dc; V_{RWM} = rated V_{RWM} (see 1.3); T_A = +100°C.				
Electrical end-points		See table I, subgroup 2				
Subgroup 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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* TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Thermal shock (glass strain)	1056	500 cycles, condition C, -65°C to +200°C.	
Temperature cycling	1051	500 cycles, condition C, -65°C to +200°C.	
Hermetic seal gross leak	1071		
Electrical measurement		See table I, subgroup 2	
Subgroup 2			22 devices c = 0
Blocking life	1048	T_A = +150°C; t = 1,000 hours +65, -0 hours; dc = 80 - 85 percent rated V_R	
Electrical measurement		See table I, subgroup 2.	
* Subgroup 3			
Not applicable			
Subgroup 4			22 devices c = 0
Thermal resistance, junction to lead	3101	Limits: 38°C/W, see 4.5.5	
Thermal resistance, junction to end caps	3101	Limits: 7°C/W, see 4.5.5	
Subgroup 5			22 devices c = 0
Barometric pressure (reduced) 1/	1001	1N5616 and 1N5618 at 8 mm Hg 1N5620 and 1N5622 at 33 mm Hg	
* Subgroup 6			22 devices c = 0
Forward surge	4066	I_{FSM} = rated (see col. 5 of 1.3.); 10 surges of 8.3 ms each at 1 minute intervals superimposed on $I_O = I_{O2}$ rated (see col. 4 of 1.3); V_{RWM} = rated (see col. 3 of 1.3.); T_A = + 25°C.	
Electrical measurement		See table I, subgroup 2.	
* Subgroup 7			
Peak reverse power		See 4.5.5 and figure 5 herein. Peak reverse power (P_{RM})= shall be characterized by the supplier and this data shall be available to the Government. Test shall be performed on each sublot.	
Electrical measurement		During the P_{RM} test, the voltage (V_{BR}) shall be monitored to verify it has not collapsed. Any collapse in V_{BR} during or after the P_{RM} test or rise in leakage current (I_R) after the test that exceeds I_{R1} in Group A shall be considered a failure to that level of applied P_{RM}. Progressively higher levels of P_{RM} shall be applied until failure occurs on all devices within the chosen sample size to characterize each sublot.	

1/ The sample size for this step stress requirement shall be determined by the supplier. A statistically significant sample size is required.

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TABLE III. Groups B, and C delta measurements. 1/ 2/ 3/ 4/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Reverse current leakage change	4016	DC method	ΔI_{R1} 4/		For JANTX and JANTXV, ≤ 250 nA dc or 100 percent, whichever is greater; for JANS, ≤ 100 nA dc or 100 percent, whichever is greater.	
2.	Forward voltage change	4011	I_F = 3 A dc; pulsed (see 4.5.1)	ΔV_{F1} 4/		± 50 mV dc maximum change from previous measured value.	

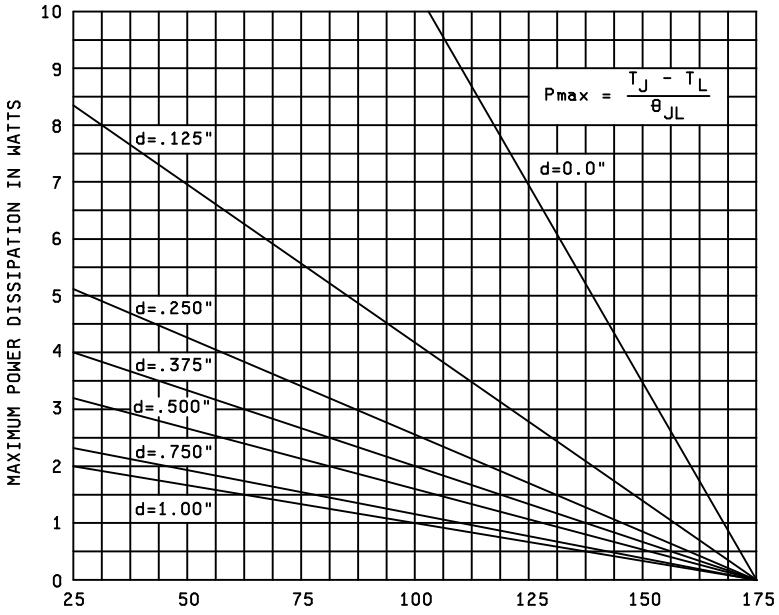
1/ The delta measurements for table VIa (JANS) of MIL-PRF-19500, appendix E, of are as follows:
   a. Subgroup 3, see table III herein, step 2.
   b. Subgroup 4, see table III herein, step 2.
   c. Subgroup 5, see table III herein, steps 1 and 2.

2/ The delta measurements for table VIb (JANTX and JANTXV) of MIL-PRF-19500, appendix E are as follows:
   a. Subgroup 3, see table III herein, step 1.
   b. Subgroup 6, see table III herein, step 1.

3/ The delta measurements for table VII of MIL-PRF-19500, appendix E, are as follows:
   a. Subgroup 2, see table III herein, steps 1 and 2 (JANS).
   b. Subgroup 6, see table III herein, steps 1 and 2 (JANS), step 1 (JANTX and JANTXV).

4/ Devices which exceed the group A limits for this test shall not be accepted.

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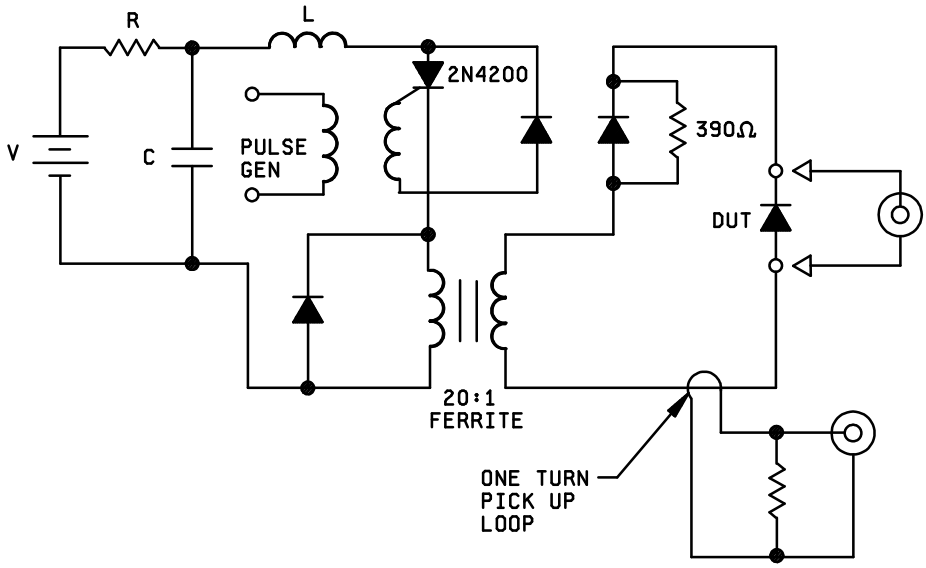
Maximum lead temperature in °C (T_L at point d from body (see table below)).
For maximum operating junction temperature of +175° with equal two-lead conditions.

d (see notes 1 and 2)	$R_{\theta JL}$ (see note 3)
Dimensions	°C/W
.000 (0.00 mm)	7
.125 (3.18 mm)	18
.250 (6.35 mm)	29
.375 (9.53 mm)	38
.50 (12.7 mm)	45
.75 (19.1 mm)	62
1.00 (25.4 mm)	76

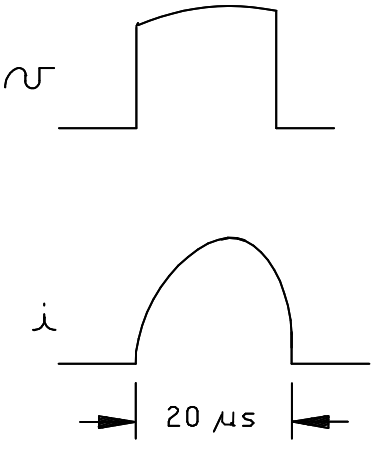
1. Lead distance point d and thermal resistance data.
2. Thermal resistance test performed at .375 inches (9.53 mm).
3. For US suffix devices: $R_{\theta JEC}$ = 7.0°C/W.

FIGURE 7. Power derating curves maximum power in watts versus lead temperature.

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NOTES:
1. L - 13T #22 pm 1 inch (25.4 mm) diameter form (air core).
2. C - 1 to 10µfd to give 20 µs pulse width.
3. V - adjustable to 200 volts for power desired in device under test.



TYPICAL WAVE FORMS

FIGURE 8. Peak reverse power measurement circuit and waveforms.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.2.1).

c. The lead finish as specified (see 3.4.1).

d. Type designation and quality assurance level.

e. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturer’s List QML No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC-VQE, P.O. Box 3990, Columbus, OH 43216-5000.

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6.4 Suppliers of die. The qualified die suppliers with the applicable letter version (e.g., JANHCA1N5614) will be identified on the QML.

JANHC and JANKC ordering information		
PIN	Manufacturer	
	14552	33178
1N5614	JANHCA1N5614 JANKCA1N5614	JANHCB1N5614 JANKCB1N5614
1N5616	JANHCA1N5616 JANKCA1N5616	JANHCB1N5616 JANKCB1N5616
1N5618	JANHCA1N5618 JANKCA1N5618	JANHCB1N5618 JANKCB1N5618
1N5620	JANHCA1N5620 JANKCA1N5620	JANHCB1N5620 JANKCB1N5620
1N5622	JANHCA1N5622 JANKCA1N5622	JANHCB1N5622 JANKCB1N5622

6.5 Supersession data. The following MIL-PRF-19500/427 types supersede MIL-PRF-19500/228, MIL-PRF-19500/286, and MIL-PRF-19500/365 types, which are inactive for new design:

Preferred	Replaces		
427	228	286	365
1N5614	1N3611	1N4245	1N4383
1N5616	1N3612	1N4246	1N4384
1N5618	1N3613	1N4247	1N4385
1N5620	1N3614	1N4248	1N4585
1N5622	1N3957	1N4249	1N4586

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA – NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/427J	2. DOCUMENT DATE 30 December 2002

3. **DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, TYPES 1N5614, 1N5616, 1N5618, 1N5620, 1N5622, 1N5614EG1, 1N5616EG1, 1N5618EG1, 1N5620EG1, 1N5622EG1, 1N5614UL, 1N5616UL, 1N5618UL, 1N5620UL, 1N5622UL, 1N5614US, 1N5616US, 1N5618US, 1N5620US, 1N5622US, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial: 614-692-0510 DSN: 850-0510 FAX: 614-692-6939 EMAIL: alan.barone@dscc.dla.mil
c. ADDRESS Defense Supply Center, Columbus ATTN: DSCC-VAC, P.O. Box 3990 Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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